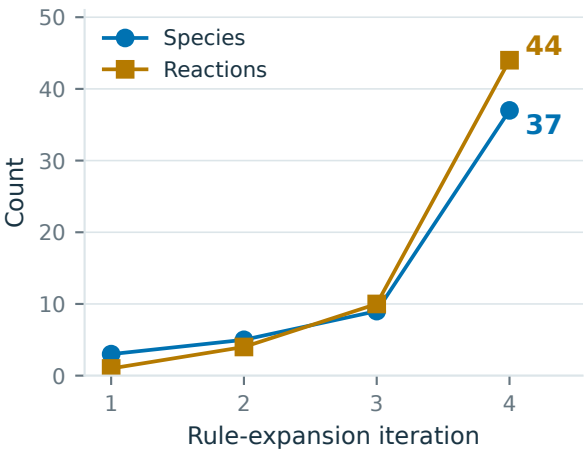
A A selected route from C_2 to C_6

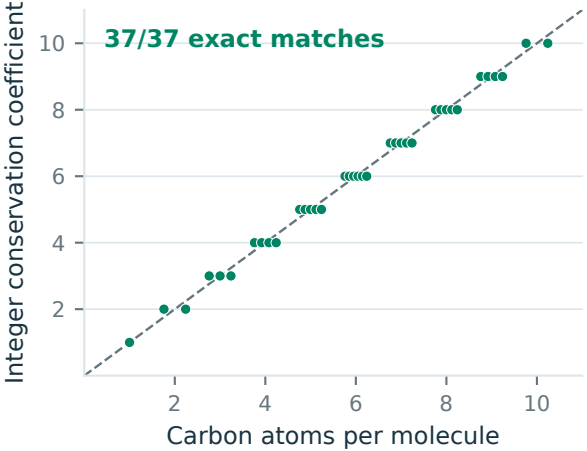
$r_1 \times 2$

2 × $HOCH_2$–CHO → **2 × HO–CH=CH–OH** two enediol donors

B Network growth



C Carbon conservation



D Equal carbon, different precursor supply

Initial inventory		Total C	Tetrose route	Hexose route
● ●●	1 + 1	3	✗	✗
● ● ●●	2 + 1	4	✗	✗
● ● ● ● ●●	4 + 1	6	✗	✗
● ● ●● ●●	2 + 2	6	✓	✗
●● ●● ●●	0 + 3	6	✓	✓

● C_1 formaldehyde ●—● C_2 glycolaldehyde ✓ certificate × selected route unrealizable